Shijie Xiao

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A highly motivated ECE graduate student with a strong foundation in computer science fundamentals, software development, and distributed systems. Proficient in building scalable, real-time systems using programming languages (Python, C++, Java), and experienced in solving complex problems through algorithmic thinking and model-driven approaches.

EDUCATION

Georgia Institute of Technology, Atlanta, GA May 2026
*Master of Science in **Electrical and Computer Engineering***
Relevant Coursework: **Computer Vision, Adv Prog Techniques**, Generate and Geometric DL, ML for Neural/Behavioral Data

Huazhong University of Science and Technology, Wuhan, China June 2024
*Bachelor of Science in **Artificial Intelligence & Automation*** GPA: **3.95**/4.00
Relevant Coursework: **Data Structures and Algorithms, Machine Learning, Principles of Computer Composition**, Linear Algebra, Calculus, Probability and Statistics

SKILLS

Programming Python, C++&C, Java, Bash, MATLAB, Linux **AI Frameworks** PyTorch, Tensorflow, OpenCV
Language Mandarin, English **Version Control** Git, Excel, SQL **Model** Diffusion Model, Transformer, CNN, LSTM

PROJECTS

Camera-based Subway Congestion Prevention System June 2023
- Developed a subway pedestrian detection and posture recognition system using PyTorch and OpenCV
- Collected and annotated a custom dataset of surveillance frames captured from Wuhan Metro stations.
- Implemented a real-time detection pipeline capable of identifying abnormal congestion patterns based on pose clustering and crowd flow estimation.

eSports Assistance System Based on Gaze Behavior Detection April 2023
- Designed and developed a personalized eSports assistant using deep learning for gaze tracking across video frames.
- Built a cross-frame gaze estimation model using convolutional neural networks and temporal feature alignment.
- Proposed a novel few-shot learning framework allowing for rapid adaptation to users with minimal calibration data.
- Final system supports real-time inference and operates with <50ms latency on standard GPUs.

Health Management System with Health Code and Trip Card May 2021
- Developed a C-based application of health code system, including QR code generation, personal data management, and epidemic risk level updates.
- Designed a structured user database supporting registration, trip tracking, and automated status updates based on user input and travel history.
- Implemented modular system features such as location-based risk evaluation and dynamic color coding.

EXPERIENCE

Research Assistant at BrainML Laboratory, Georgia Tech Dec 2025
- Conducted research on machine learning methods for trajectory analysis to understand animal behavior patterns.
- Implemented algorithms to predict mice movement and modeled spatiotemporal interactions using graph structures.
- Developed training pipelines and introduced a novel geodesic loss function that improved performance.

Research Assistant at The Hong Kong University of Science and Technology, Hong Kong April 2024
- Conducted research on applying noisy label segmentation to diffusion models for image generation tasks.
- Proposed a novel approach for learning from noise-infused labels in both unsupervised and semi-supervised settings.
- Utilized unlabeled and noisy-labeled image datasets to improve segmentation performance using Stable Diffusion.

Research Assistant at The Chinese University of Hong Kong, Hong Kong Mar 2023
- Developed a DDPM-based denoising method for speckle noise in OCT medical images to improve diagnostic accuracy.
- Implemented noise modeling, diffusion training pipelines, and enhanced convergence speed and prediction efficiency.
- Reproduced baseline models including DN-CNN, U-Net, BM3D, TCFL-OCT, and DDPM for comparative evaluation.
- Wrote Python scripts to compute denoising metrics such as PSNR, SSIM, and GCMSE.

Participant at CVPR 2023 BDD100K Challenges Jun 2023
- Developed a multi-object tracking pipeline for autonomous driving using state-of-the-art machine learning models.
- Implemented object detection with advanced detectors including YOLO-NAS, YOLOv8, and DINO.
- Fine-tuned models on the MOT20 dataset and applied ByteTrack and BoT-SORT for object tracking.
- Ranked in the top 20% for Multi-Object Tracking (MOT) and achieved first place in Object Detection (OD) tasks.

Winter School at North Carolina State University, Raleigh, NC Feb 2023
- Conducted research on motion classification of brain-computer interface (BCI) signals using machine learning methods.
- Developed EALNet, a custom BCI classification model combining EEGNet, LSTM, and Transformer architecture.
- Achieved 10% improvement in classification accuracy on standard BCI Competition datasets.

Undergraduate Research Assistant at Huazhong University of Science and Technology, Wuhan, China Dec 2022
- Conducted research on text detection of musical notation using Optical Character Recognition (OCR) techniques.
- Built and annotated a specialized dataset from online musical tune data for model training and evaluation.
- Evaluated models including DBNet, DBNet++, and PANet using the MMOCR framework and detection libraries.